GEOG 4/5/7 9073: Environmental Analysis in R

Week 8.01: Geometry, data structures, and the flipped classroom

Dr. Bitterman

Today's schedule

- Open discussion
- Something different - you get to choose
- For next class

Anything to discuss? Questions?

Today's options

Required

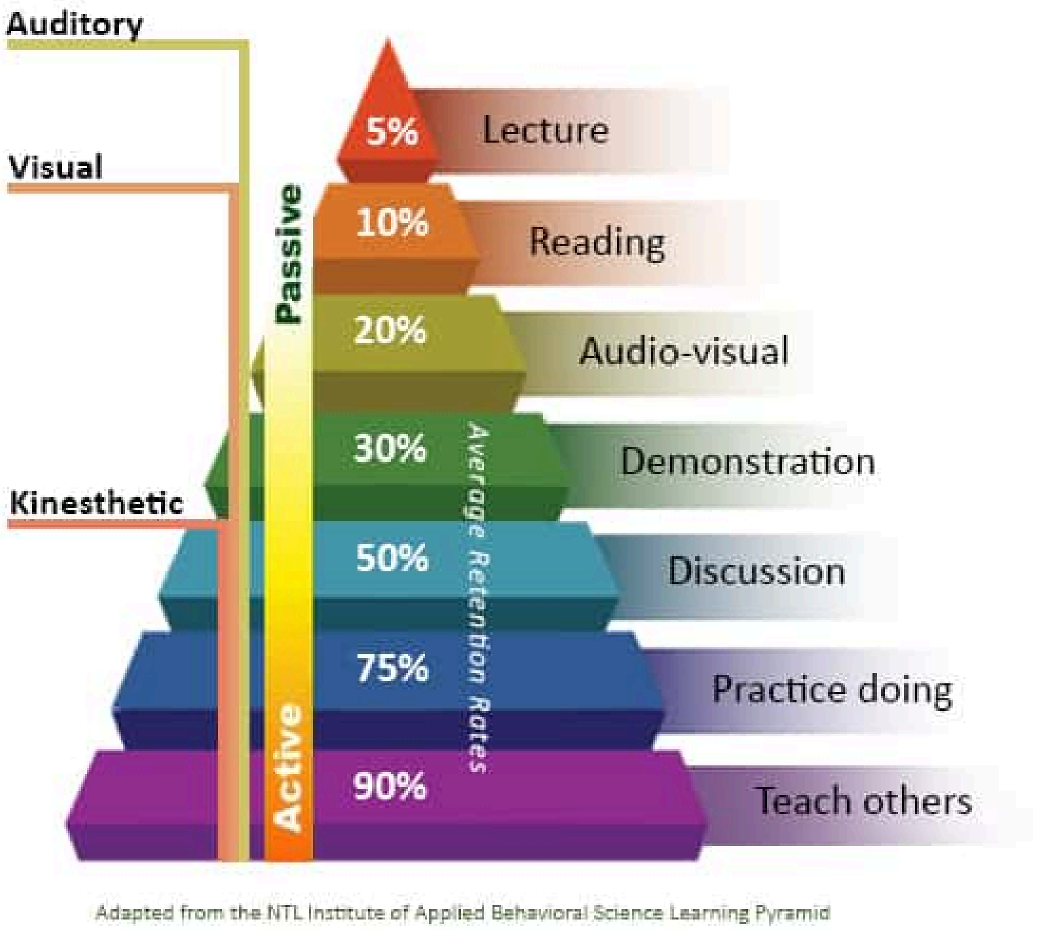
1. Code reading exercise

As a class, let's pick one of the following:

2. The tasks from 4.02
3. Chapter 5 "lessons"

Let's talk about the options

How do we (and you) learn best? Thoughts?



(to be fair, there are counterarguments to this model)

This week's activity

- You all read (or should have) Chapter 5 from Lovelace (https://geocompr.robinlovelace.net/geometric-operations.html)
- Instead of me providing you with a step-by-step walkthrough of the readings, **you're** going to do the teaching
- A quasi-"flipped classroom"

What to do

- Form small groups (I've assigned the groups)
- Each group will be assigned a topic (or topics) from this week's readings
- Your tasks:
    - Develop a short (~10 min) lesson demonstrating the method(s)
    - Include:
        a. Learning objectives (what students will learn)
        b. Why the concepts/methods are important/relevant
        c. How a student would accomplish the task(s)
        d. A way to check for learning (and teaching != learning)

All relevant resources can be found in the Lovelace chapter, but use what you think is relevant

What you can use

- Anything
    - Web resources
    - Sample data
    - Whatever format you want (e.g., PowerPoint, R Markdown, something else)

Tasks and teams

Creating geometry, binary transformations, type transformations

- NAMES

Simplify, scale, shift, and rotate geometry

- NAMES

Raster aggregation, disaggregation, and resampling

- NAMES

Choice?

Required code reading activity

It's important that our code be readable, reproducible

- In group/collaborative settings, you'll have to read the code of others
- So today, you'll read some of MY code
- And interpret it

About the code

- It contains functions AND procedural code that uses those functions
- It WAS a bit old, but I have updated it to reflect new packages and paradigms
- It relies on data that I have added to `/data/erie_cicyano/`

About the GitHub repository

- Let's talk strategies
- Clone vs. fork vs. downloading the repo vs. downloading only what you need

Your task

- In the "/src" directory of the course repo, find the "code_reading_ex.R" file
- I've stripped most of the comments from the document
- Your tasks...

    i. look at the code (and comments) to understand the "flow" of the code
    ii. then, go through the code, line-by-line
    iii. think about what the code is doing, then test it
    iv. fill in the comments (behind the "#") with what the code does

- You can run it at any time up to any step
- I'm assigning teams

Questions?

For next week

1. Proposals due MONDAY
2. Next week is spatial autocorrelation